

Day 21 Student Handout

Geometry Summer Credit | Surface Area, Volume, and Cross-Sections

Name:	Date:	Period:
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Learning Targets

I can find surface area and volume of common solids. I can explain the difference between surface area and volume. I can describe cross-sections of prisms, cylinders, pyramids, cones, and spheres.

Lesson 21A: Prisms and Cylinders

Warm-Up

1.	Find the area of a rectangle with length 12 and width 7.
2.	Find the area of a circle with radius 5. Leave the answer in terms of pi.
3.	A rectangular prism has length 8, width 5, and height 3. How many unit cubes fit inside it?

Big Difference

Surface area measures the outside covering of a solid in square units. Volume measures the space inside a solid in cubic units.

Problem	Work / notes
A rectangular prism has length 10, width 6, and height 4. Find volume and surface area.	Volume: Surface area: Units:
A cylinder has radius 5 and height 12. Find volume in terms of pi.	Base area: Volume:
A cylinder has radius 4 and height 9. Find surface area in terms of pi.	Two bases: Lateral area: Total surface area:

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Lesson 21B | Pyramids, Cones, Spheres, and Cross-Sections

Formula Selection

Common Volume Formulas

Prism or cylinder: $V = Bh$. Pyramid or cone: $V = \frac{1}{3}Bh$. Sphere: $V = \frac{4}{3}\pi r^3$. For surface area, think about all outside faces or curved surfaces.

Problem	Work / notes
A square pyramid has base side 9 and height 12. Find volume.	Base area: Volume:
A cone has radius 6 and height 10. Find volume in terms of pi.	Base area: Volume:
A sphere has radius 3. Find volume and surface area.	Volume: Surface area:

Cross-Sections

Slicing Solids

A cross-section is the two-dimensional shape made when a plane slices through a solid. A horizontal slice of a cylinder is a circle. A vertical slice through the center of a cylinder is a rectangle. A horizontal slice of a cone is a circle.

4.	Describe one possible cross-section of a rectangular prism and one possible cross-section of a cone.
5.	Which is more appropriate for paint: surface area or volume? Which is more appropriate for filling a tank?